10 **(a)** Define *half-life*.

.....

.....

..... [1]

(b) The isotope ^{59}Fe is a $\beta-$emitter with a half-life of 45 days. In a medical investigation of blood disorders, a dose of ^{59}Fe with an initial activity of 8.5×10^{7} Bq is injected into the blood stream of a patient.

(i) Calculate

1. the probability per second of the decay of a ^{59}Fe nucleus,

probability = ... s^{-1} [2]

2. the initial number of ^{59}Fe nuclei injected into the body,

number = ... [2]

3. the time taken for the activity to be one-fifth of the initial value.

time = ... s [2]

(ii) Explain why in practice, the time taken for the activity in the patient to become one-fifth of the initial value is shorter than that calculated in **(i)3**.

.....

..... [1]

(c) (i) On Fig. 10.1, sketch a graph to show the variation with nucleon number A of the binding energy per nucleon B_E. Label in the graph an approximate value, in MeV, of the maximum binding energy per nucleon.

Fig. 10.1

[2]

(ii) Using Fig. 10.1, explain why energy is released in a fusion reaction.

.....

.....

..... [2]

(iii) A possible nuclear reaction which has been suggested for the generation of energy is represented by the following equation:

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + X$$

Data for the above reaction are given in Fig. 10.2.

nucleus	mass of nucleus / u
X	1.008665
$^{2}_{1}H$	2.014102
$^{3}_{1}H$	3.016050
$^{4}_{2}He$	4.002603

Fig. 10.2

1. Identify the nucleus represented by the symbol *X*.

..... [1]

2. Using the data in Fig. 10.2, determine the energy produced by 150 kg of an appropriate mixture of the hydrogen isotopes.

energy = ... J [4]

3. The average power needed by a nuclear submarine is 110 MW and the efficiency of the above process is 15%.

Determine the length of time, in years, that 150 kg of the mixture can supply sufficient energy to the submarine.

time = ... years [3]

End of paper